

END-TO-END MACHINE LEARNING MASTER COURSE

Master Study Notes: Lessons 1 to 6

Project: Universal Customer Churn & Retention Prediction System

Lesson 1 & 2: ML Foundations & Business Problem Mapping

Machine Learning learns rules dynamically from historical data (X , y), unlike traditional programming which relies on hardcoded human rules.

Customer Churn is the universal problem of customers leaving a business. Retaining existing customers costs 5x to 7x less than acquiring new ones.

Lesson 3 & 4: Dataset Anatomy (X vs y) & Data Types

- **Features (X):** Input variables (Tenure, Bill, Age).
- **Target (y):** Output label (Churn: 1 or 0).
- **Numerical Data:** Numbers used in math (Continuous/Discrete).
- **Categorical Data:** Text labels/groups (Nominal/Ordinal).

Lesson 5: Pandas Inspection & Data Cleaning

Using `df.info()`, we detected 20 missing values in `Total_Charges_INR`.

💡 Median Imputation Rule

For monetary features sensitive to extreme outliers, missing values are filled with the **Median (50th percentile)** instead of the Mean.

Lesson 6: Categorical Encoding (Text to Numbers)

Using `pd.get_dummies(df, drop_first=True)`, text categories are converted into binary True/False (1/0) columns, avoiding false numeric rankings and avoiding the Dummy Variable Trap.

Clean Master Python Script (index.py)

```
import pandas as pd

# Load Data
df = pd.read_csv('data/customer_churn_data.csv')

# Impute Missing Values with Median
df['Total_Charges_INR'] = df['Total_Charges_INR'].fillna(df['Total_Charges_INR'].median())

# One-Hot Encoding
df_encoded = pd.get_dummies(df, columns=['Gender', 'Contract_Type', 'Payment_Method', 'Tech_Support'])

print("Preprocessed Data Shape:", df_encoded.shape)
print(df_encoded.head())
```